Azure AI & Microsoft Foundry

Comprehensive Compilation of Exam Scenarios and Practice Questions

Drag and Drop Code Completion

Scenario 1: Forcing Tool Execution in Ticket Triage

You have a Microsoft Foundry project that contains a deployed ticket-triage agent. You discover that sometimes the agent responds without calling any tools, even when a tool is required. You need to ensure that the agent calls a tool during execution. How should you complete the Python code payload?

Target Key Box	Target Value Box
`[ Select Target Key ]`	`[ Select Target Value ]`

```
run_payload = {
    "assistant_id": agent_id,
    [Target Key Box]: [Target Value Box],
    "metadata": {
        "scenario": "ticket-triage"
    }
}
```

A. Target Key: `"tool_choice"` | Target Value: `"auto"`

B. Target Key: `"tool_choice"` | Target Value: `"required"`

C. Target Key: `"tools"` | Target Value: `"required"`

D. Target Key: `"response_format"` | Target Value: `"type"`

Dropdown Selection

Scenario 2: CI/CD Workflow Evaluation & Branch Protection

You have a Microsoft Foundry project that contains an agent. You use a GitHub Actions workflow for CI/CD. You need to configure the workflow to automatically evaluate the agent when a pull request (PR) is created and prevent branches from merging if the evaluation results do NOT meet the defined thresholds. How should you configure the workflow?

Configuration Property	Selected Option
Authentication method	`[ Select Option ]`
If the evaluation results are NOT met, configure the workflow to	`[ Select Option ]`

A. Authentication: A static API key stored in repository variables | Action: Warn

B. Authentication: An Azure Login action that uses OpenID Connect (OIDC) | Action: Warn

C. Authentication: An Azure Login action that uses OpenID Connect (OIDC) | Action: Fail

D. Authentication: Anonymous access to the Foundry project endpoint | Action: Fail

Multiple Choice

Scenario 3: Real-Time Audio Chat Turn-Taking Dynamic

You are creating an agent workflow in a Microsoft Foundry project to support natural voice interactions. The agent must receive continuous audio input, convert the input into text for reasoning, and then return spoken responses to a user. The workflow must support turn-taking dynamics where the agent begins to generate speech output before the user finishes speaking, operating with low latency. What should you do?

A. Use speech translation to convert the audio into another language and return the translated text.

B. Use real-time speech to text for incoming audio and text to speech for agent responses.

C. Use an embeddings model to encode the audio, and then decode the audio into text and speech.

D. Use batch transcription to convert the audio input and return text responses from the agent.

Multiple Choice

Scenario 4: Live Call Transcription Monitor

You are building a speech processing solution in Microsoft Foundry for a customer support platform. The platform will transcribe live phone calls so that supervisors can view transcripts and detect issues while the calls are in progress. The call audio arrives as a continuous stream from the telephony system. You need to ensure that transcripts appear within only a few seconds of the audio stream. What should you do?

A. Use speech translation to generate the transcripts into multiple languages.

B. Use text to speech by using a custom neural voice.

C. Run a batch transcription job on recorded audio files.

D. Use real-time speech to text to process streaming audio input.

Multiple Choice

Scenario 5: Fine-tuned Speech Model REST API Authorization Error

You have a Microsoft Foundry project that contains an agent. The agent uses Azure Speech in Foundry Tools. You fine-tune a baseline speech to text model for the en-US locale and publish the model. The agent calls the Speech to text REST API and returns an error message indicating that the project ID is invalid. You need to set the project property to the correct ID. To what should you set the project property?

A. the project URL

B. the custom speech project ID

C. the project ID

D. the custom speech endpoint URL

Multiple Choice

Scenario 6: Cost Observability and Analysis

You have a Microsoft Foundry project that contains a high-traffic agent. After a recent update, operational costs increase significantly. Monitoring confirms that the volume of user traffic remains unchanged. You suspect that changes to the request or response characteristics are causing the increase. You need to identify whether additional costs are driven by model input size, model output size, or expanded tool usage. Which observability capability should you use?

A. evaluation metrics

B. run success rate

C. latency

D. token usage

Multiple Choice

Scenario 7: Least Privilege Inference Authorization (RBAC)

You have a Microsoft Foundry project that contains a model deployment. You have an application that calls the deployment by using the Azure OpenAI v1 API and DefaultAzureCredential. The developers receive HTTP 403 errors when they send inference requests, even after running 'az login'. You need to ensure the developers can perform model inference following the principle of least privilege. Which role-based access control (RBAC) role should you assign?

A. Contributor

B. Cognitive Services Data Reader

C. Cognitive Services OpenAI User

D. Cognitive Services User

Multiple Choice

Scenario 8: Deterministic Multi-Agent Orchestration

You have a Microsoft Foundry project that contains three agents: TriageAgent (classifies requests), PolicyAgent (searches content), and ActionAgent (calls APIs). You need to orchestrate the agents to ensure customer requests meet requirements: support a deterministic, step-based process using conditional branching, shared state across agents, and optionally trigger a ticket action based on triage. The solution must minimize development effort. What should you include?

A. a workflow

B. a multi-agent group chat session

C. separate agent runs coordinated in the application code

D. threads and runs without a workflow

Multiple Choice

Scenario 9: Multi-Turn Parallel Vector Index Ingestion (RAG)

You have a chat app in a Microsoft Foundry project and an Azure AI Search vectorized index. You need to connect to the index to meet the requirements: complex questions must retrieve information from multiple chunks, multi-turn conversations must influence retrieval planning, and retrievals must run in parallel to reduce latency. Which retrieval approach should you use?

A. iterative retrieval

B. classic Retrieval Augmented Generation (RAG)

C. agentic Retrieval Augmented Generation (RAG)

D. chain of thought

Multiple Choice (Select Two)

Scenario 10: External App Tracing Architecture

You have a Microsoft Foundry project that contains a prompt agent used by a customer support web app. The agent is invoked from a Python service that does NOT run in the Foundry portal. You need to implement end-to-end tracing to capture latency breakdowns and exceptions across agent runs. Which two components can you use? (Each correct answer presents a complete solution).

A. a Log Analytics workspace

B. Microsoft Sentinel

C. Application Insights

D. OpenTelemetry

E. the Azure Monitor Agent

Dropdown SDK Integration

Scenario 11: Python Managed Identity Token Generation

You have a Python application named App1 that integrates with a Microsoft Foundry project named Project1. You need to ensure that App1 authenticates by using a Microsoft Entra managed identity and sends prompts to a deployed model by using the Azure OpenAI Responses API. How should you complete the code?

Code Location Target	Code Selection
credential = [Dropdown_1]()	`[ Select Dropdown_1 ]`
response = openai_client.responses. [Dropdown_2](	`[ Select Dropdown_2 ]`

A. Dropdown_1: `AzureKeyCredential` | Dropdown_2: `create`

B. Dropdown_1: `DefaultAzureCredential` | Dropdown_2: `create`

C. Dropdown_1: `DefaultAzureCredential` | Dropdown_2: `retrieve`

D. Dropdown_1: `ClientSecretCredential` | Dropdown_2: `compact`

Dropdown Model Execution Parameters

Scenario 12: Deterministic Consistency & Reasoning Quality

You have a Microsoft Foundry project containing a deployed chat model. You have a Python service that sends API requests to the model, integrated with an automated validation system comparing outputs against approved patterns. Wording differences cause validation mismatches. You need to update parameters to stabilize outputs and maximize reasoning quality using a model configured with an active thinking block.

Hyperparameter Field	Target Value
temperature =	`[ Select Value ]`
output_config = {"effort":	`[ Select Value ]`

A. `temperature = 1` | `output_config = {"effort": "high"}`

B. `temperature = 0` | `output_config = {"effort": "low"}`

C. `temperature = 0` | `output_config = {"effort": "high"}`

D. `temperature = 2` | `output_config = {"effort": "medium"}`

Multiple Choice

Scenario 13: Mixed Request Complexity Cost Optimization (Cascading)

You have a Microsoft Foundry project that serves a high-volume chat app. Most requests are simple FAQs, but some require advanced reasoning. You need to reduce costs and latency for common queries, without degrading the quality of the responses to complex questions. What should you do?

A. Increase the value of the max_tokens parameter for all the requests.

B. Route all the requests to a smaller model.

C. Use a model cascade that routes the requests to different models.

D. Route all the requests to the most capable model.

Multiple Choice

Scenario 14: QR and Table Document Layout Extraction

You have a Microsoft Foundry project that contains an agent. The agent ingests scanned PDF vendor invoices that contain tables and embedded QR codes. The agent must preserve the PDF layout in the extracted output to ensure that downstream processing can reference sections and tables. You plan to call Azure Content Understanding in Foundry Tools. You need to extract content/layout elements and detect QR codes without requiring a language model deployment. Which built-in analyzer should you use?

A. prebuilt-read

B. prebuilt-layout

C. prebuilt-documentSearch

D. prebuilt-documentFieldSchema

Multiple Choice

Scenario 15: Structure-Aware RAG Table Ingestion

You have a Microsoft Foundry project that contains an agent using a knowledge source built from documents in Azure Blob Storage (digitally scanned PDFs with multi-page tables). Your current ingestion job extracts only plain text, causing loss of table structure, headings, and page metadata. Users need to retrieve specific table rows across pages. You need to configure a RAG ingestion job that performs OCR, preserves hierarchies, and tracks page metadata. What should you do?

A. Use advanced data parsing to reingest the documents.

B. Use OCR and page-level chunking.

C. Use basic parsing and fixed-size chunking.

D. Use page-level OCR extraction and store each page as a single chunk.

Multiple Choice

Scenario 16: Portal Markdown Ingestion Configuration

You have a Microsoft Foundry project named Project1 that contains an application that processes PDF vendor invoices. You need to configure Azure Document Intelligence in Foundry Tools to generate a Markdown output that preserves the sections and table structure of the PDFs. The solution must minimize development effort. What should you do?

A. Configure content=markdown when you analyze the document.

B. Configure output=figures when you analyze the PDF.

C. Increase the confidence threshold.

D. Set the output_content_format=ContentFormat.MARKDOWN value.

Multiple Choice

Scenario 17: Multi-Column Layout Extraction Sequence

You have a Microsoft Foundry project that contains an agent. You need to process mixed-format documents that contain scanned text, tables, and multicolumn layouts. The extracted content must preserve the document structure and be converted into the Markdown format for downstream reasoning. What should you configure first?

A. an Azure Language in Foundry Tools text analysis model deployment

B. a generative chat completion request

C. an Azure Content Understanding in Foundry Tools analyzer

D. an Azure OpenAI Responses API call that uses a multimodal model

Multiple Choice

Scenario 18: Target-Locking Specific MCP Framework Tools

You have a Microsoft Foundry project that contains an agent. The agent has a Model Context Protocol (MCP) tool that queries a knowledge base stored in Azure AI Search. Some agent runs return answers from the base model without invoking the knowledge base, which results in responses without grounded citations. You need to add the correct tool_choice parameter to the code to deterministically force the agent to invoke the MCP tool on each run. What should you add?

A. tool_choice={"required"}

B. tool_choice={"auto"}

C. tool_choice={"type": "mcp"}

D. tool_choice={"type": "knowledge_base"}

True / No Matrix

Scenario 19: Image Security Vector Evaluation (Case A)

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses. You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model. You need to implement controls to mitigate the risk.

Proposed Solution: You configure a prompt shield for documents.

Does this meet the goal?

A. Yes

B. No

True / No Matrix

Scenario 20: Image Security Vector Evaluation (Case B)

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses. You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model. You need to implement controls to mitigate the risk.

Proposed Solution: You configure image moderation to block unsafe content before processing the images.

Does this meet the goal?

A. Yes

B. No

True / No Matrix

Scenario 21: Image Security Vector Evaluation (Case C)

You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses. You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model. You need to implement controls to mitigate the risk.

Proposed Solution: You configure a prompt shield for user prompts.

Does this meet the goal?

A. Yes

B. No

True / No Matrix

Scenario 22: Regulatory Completeness (Case A - Reflection)

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents. Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content. You need to improve response completeness.

Proposed Solution: You add a reflection pass that regenerates the response if the required clauses are missing.

Does this meet the goal?

A. Yes

B. No

True / No Matrix

Scenario 23: Regulatory Completeness (Case B - Max Tokens)

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents. Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content. You need to improve response completeness.

Proposed Solution: You increase the value of the max_tokens parameter.

Does this meet the goal?

A. Yes

B. No

True / No Matrix

Scenario 24: Regulatory Completeness (Case C - Blocker Evaluation)

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents. Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content. You need to improve response completeness.

Proposed Solution: You run an evaluation flow that scores responses for completeness and blocks responses that fall below a defined threshold.

Does this meet the goal?

A. Yes

B. No

True / No Matrix

Scenario 25: Regulatory Completeness (Case D - Temperature)

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents. Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content. You need to improve response completeness.

Proposed Solution: You increase the value of the temperature parameter.

Does this meet the goal?

A. Yes

B. No